Rajeev Jain

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Principal Specialist, Research Software Engineering | AI Systems and Verification | HPC | Scientific Computing

Profile

Principal research software engineer, at Argonne since 2009, turning research prototypes into production systems across AI infrastructure, scientific computing, and HPC. Lead developer of UXarray, a Python library for unstructured climate-grid analysis used at DOE labs and NCAR, including an MCP server that gives AI agents typed, scoped, provenance-tracked access to Earth-system meshes on HPC clusters via Globus Compute — the alternative to handing a team general-purpose chat and calling it a strategy. Equally focused on verifying what such systems produce: caught silent numerical drift, hidden heap allocation, and unreachable vendor throughput figures in code and benchmarks that read as correct. Ran 10,000+ cancer-AI training experiments on Summit; ported the SFNO climate emulator to Intel XPUs on Aurora. Two R&D 100 Awards (CANDLE 2023, FLASH-X 2022).

Professional Experience

Argonne National Laboratory — **2009 – Present**
Principal Specialist, Research Software Engineering — *Mathematics and Computer Science Division, Chicago, IL*

- **UXarray and MCP server.** Lead developer of a Python library for unstructured climate grid analysis used at DOE labs and NCAR. Designed grid I/O for ESMF / MPAS / SCRIP / HEALPix and conservative zonal averaging. Built the UXarray MCP server: 20+ typed tools letting AI agents inspect, subset, visualize, and run analysis on Earth-system meshes locally or on HPC clusters (Argonne Improv, UCAR Derecho) via Globus Compute, with structured provenance. Presented at SciFM26.
- **AI code and benchmark verification.** Independent measurement applied to AI tooling and machine-assisted code. Published a reproducible harness for local LLM inference: bytes-per-token parsed from GGUF tensor tables rather than file size (which overstates 64% on gathered-embedding models), and memory bandwidth measured at 340 GB/s against an unreachable 400 GB/s datasheet pin rate — showing four different architectures all land within 36–41% of bandwidth, so throughput is set by model size and runtime, not architecture. In numerical Python, traced 0.68 m of drift on a production Earth mesh to catastrophic cancellation inside a spherical cross product — a difference of nearly equal products, not a running sum — and proposed error-free-transformation arithmetic for the kernel (merged upstream, UXarray PR #1513); recovered a 1.67× speedup by removing heap allocation a faithful C++-to-Python port had silently introduced.
- **SFNO climate emulator on Aurora.** Ported the SFNO climate emulator to Aurora's Intel XPU stack (60,000+ GPUs): PMIX/PALS launcher mapping, XPU/CUDA device branching, device-aware AMP (GradScaler on CUDA, bf16 on XPU), first stable DDP baseline. Established smoke-test and queue-aware bring-up scripts separating launcher, backend, and device-placement failures. Measured one-node baseline: 12 XPU ranks under DDP, ∼12 s steady-state epochs after a warmup epoch 8× longer.
- **CANDLE / IMPROVE.** Built CANDLE/Supervisor HPO workflow (Swift/T, mlrMBO, DEAP, Hyperopt/TPE); ran 10,000+ cancer drug response experiments on Summit, Theta, and Cori. Applied noise injection and counterfactual analysis to NT3 tumor classification; identified PLOD2, RGS5, and TP53I13 as decision-boundary drivers. Led IMPROVE: `improvelib` Python package with GitHub Actions CI/CD (unit tests, Docker, Parsl CSA) and UNO dual-branch neural network for cross-study benchmarking. 15+ researchers across Argonne, LLNL, ORNL. R&D 100 Award, 2023.
- **FLASH-X.** I/O and compression lead for a million-line multiphysics engine. Async HDF5 with Argobots plus SZ3 / ZFP compression reduced checkpoint overhead 40–70% on Summit with 50%+ storage savings. Enabled cross-checkpoint restart between AMReX and Paramesh. R&D 100 Award, 2022.
- **MeshKit and Urban ECP.** PI for MeshKit (DOE NEAMS, 2009–2016): C++ toolkit for reactor mesh generation; parallel CoreGen produced a 101M-element MONJU mesh on 712 processors in about 7 minutes. Led Urban ECP: coupled WRF/HRRR → EnergyPlus → Nek5000 workflow with full 3D downtown Chicago domain. Published in *Journal of Building Performance Simulation*, 2020.

University of Chicago — **2023 – Present**
Staff At-Large — *Consortium for Advanced Science and Engineering (CASE), Chicago, IL*

- Joint appointment supporting cancer pharmacogenomics and Earth science software collaborations; mentoring graduate students on software design and HPC workflows.

Arizona State University **2007 – 2009**
Research and Teaching Assistant *Structural and Computational Mechanics Lab, Tempe, AZ*

- FEM research on blast mitigation and design optimization; supported structural engineering coursework.

Earlier Experience **2005 – 2007**
Project Engineer, Wipro Technologies | Engineering Intern, Tata Motors *India*

Technical Skills

Languages: Python, C++, Fortran, Bash, R, SQL
ML / AI: PyTorch, TensorFlow, NumPy, Pandas, Xarray, Scikit-learn; Bayesian HPO (mlrMBO, Hyperopt/TPE), genetic algorithms (DEAP)
AI Systems: MCP server design (typed tools, scoped access, structured provenance), Claude API, LLM tool-use and agent workflows, local inference (llama.cpp, GGUF, quantization), Globus Compute for HPC-backed agents
Correctness / Perf: Numerical analysis (compensated summation, error-free transformations), roofline and bandwidth analysis, profiling, review of ported and machine-generated numerical code, reproducible benchmark design
HPC / Systems: MPI, OpenMP, HDF5, NetCDF, MOAB, Argobots, SZ3/ZFP, Intel XPU / CUDA, PBS/SLURM
Workflow / DevOps: GitHub Actions, Docker, Singularity, Parsl, Swift/T, release automation, CI/CD, Globus Compute
Domains: Climate and weather modeling, cancer pharmacogenomics, computational physics, reactor mesh generation

Selected Publications

- **Jain, R.** and Jacob, R. "Beyond Tool Execution: Evaluating Scientific MCP Interfaces with UXarray." *1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026*, Naples — accepted, to be presented. Artifacts: doi:10.5281/zenodo.21463232.
- Partin, A., ..., **Jain, R.**, et al. "Benchmarking community drug response prediction models." *Briefings in Bioinformatics*, 2026.
- **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *SC24 / DRBSD-10*, 2024.
- **Jain, R.**, Wozniak, J. M., et al. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response." *CAFCW at SC24*, 2024.
- Wozniak, J. M., **Jain, R.**, et al. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics*, 2018.

Full list: Google Scholar.

Awards, Recognition, and Service

- Program committee member, 1st Workshop on Agentic AI for Large-scale Science (AGENT4SC) at IEEE eScience 2026.
- R&D 100 Award, CANDLE (2023) | R&D 100 Award, FLASH-X (2022).
- DOE SBIR Phase I & II (RGG reactor mesh generator; Kitware, 2014–2017).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).

Education

University of Chicago 2020
M.S. in Computer Science

Arizona State University 2009
M.S. in Structural Engineering Graduate Fellowship Recipient

Indian Institute of Technology (ISM) Dhanbad 2006
B.Tech. in Mechanical Engineering